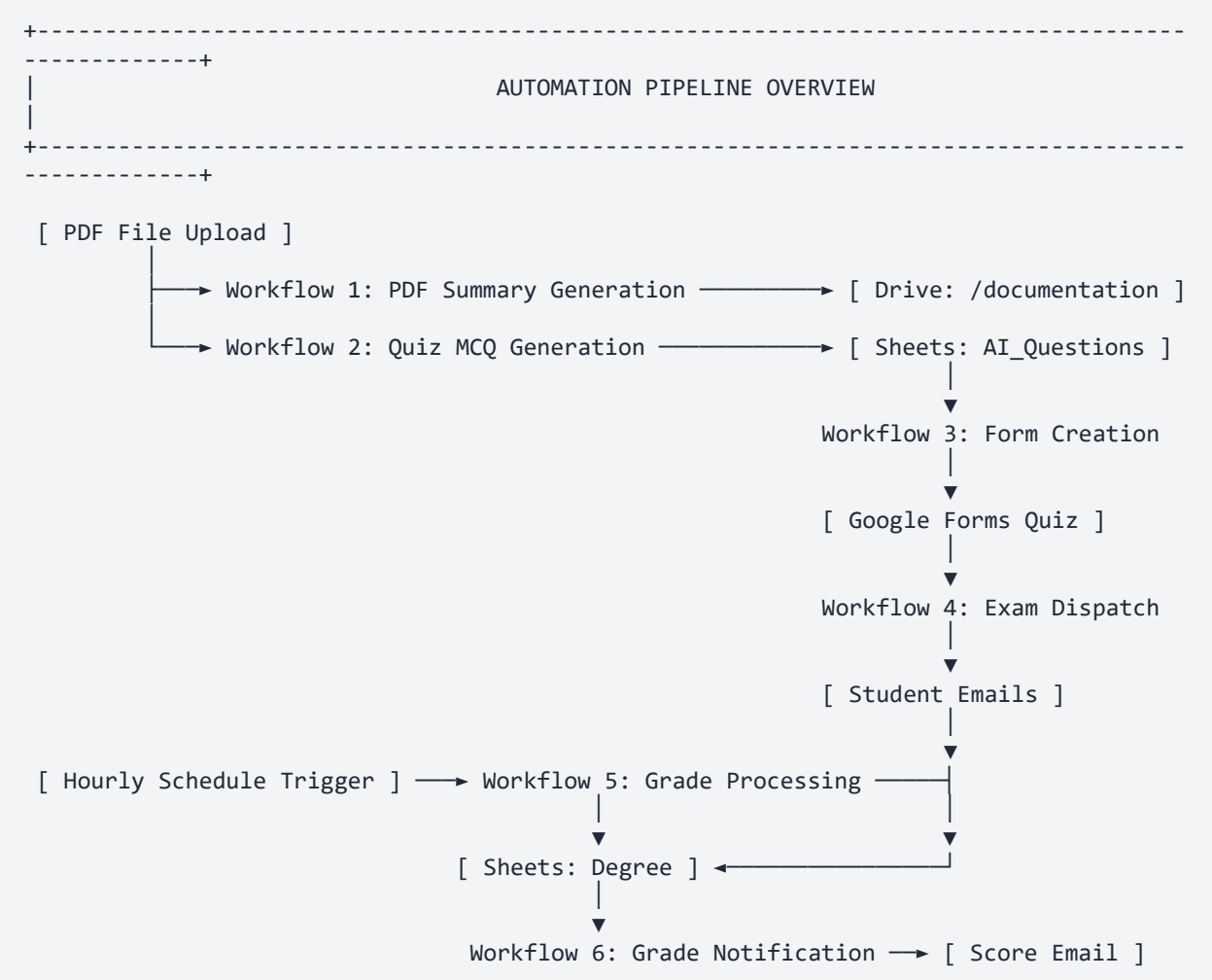
n8n Educational Automation Pipeline

Handover Documentation & Operational Technical Manual

GitHub link:	https://github.com/rafeekelgamal/edupilot
Document Title:	End-to-End n8n Educational Automation Pipeline Handover Package
Target Audience:	Systems Administrators, DevOps Engineers, and Future Workflow Maintainers
Scope:	Automation Workflows 1–6, Google Cloud/Drive/Sheets Integrations, AI Processing Pipelines, and External API Specifications
Group Member:	Dr. Ayman Abdelfattah Dr.Weaam Mohamed Amir Gerges Rafeek Magdy Martina Fakhry Zahraa Hagag

1. Executive Summary

The n8n Educational Automation Pipeline is an end-to-end cloud-integrated system designed to process educational materials, synthesize learning content, create interactive assessments, dispatch exams, evaluate student submissions, and communicate performance metrics.



Core Pipeline Capabilities

- **1. Content Ingestion & Summarization (Workflow 1):** Ingests raw educational PDFs from Google Drive, extracts machine-readable text, utilizes Google Gemini AI (models/gemini-3.5-flash-lite) to extract lesson titles, summaries, key concepts, and learning objectives, and generates structured text files in a cloud repository.
- **2. Automated Assessment Generation (Workflow 2):** Processes source PDFs to generate 30 multiple-choice questions (MCQs) per lesson using Google Gemini, formats items into structured schemas, and appends them to a central staging spreadsheet.

- **3. Form Provisioning & Quiz Construction (Workflow 3):** Reads staged questions from Google Sheets, groups items by exam name, creates a new Google Form via API, configures quiz settings, configures points and answer keys, records form metadata, and invokes downstream dispatching.
- **4. Targeted Exam Dispatch (Workflow 4):** Cross-references student registries with form registries, generates personalized HTML email templates containing direct access links, dispatches communications via the Gmail API, and updates tracking flags.
- **5. Grade Ingestion & Score Parsing (Workflow 5):** Polls active Google Form responses hourly, fetches submission metadata, normalizes student identification keys, calculates overall scores, prevents duplicate entries, and appends records to a master gradebook.
- **6. Automated Score Notification (Workflow 6):** Queries pending grade records, dispatches individual score reports to students via Gmail with error-handling logic, and updates status flags (ok or error) based on execution success.

2. System Environment Directory & Data Flow Reference

Primary Google Drive Workspace

Root Folder URL: https://drive.google.com/drive/folders/1fHuRrySnjR8_-3H8ZwpwnobBiYfORoaP

Role: Primary storage infrastructure hosting source lesson PDFs and generated AI documentation.

Sub-Directories

- **/PDF file (Primary Input Directory):** Drop folder where instructors upload raw lesson materials in PDF format. Monitored by Google Drive Trigger1 in Workflow 1 (1-create doc.json) via fileCreated events and Workflow 2 (2-create quiz.json) to fetch source files.
- **/documentation (Primary Output Directory):** Stores AI-generated lesson summaries and structured educational documents. Populated by the Create file from text node in Workflow 1 (1-create doc.json).

AI Question Repository

Spreadsheet URL: https://docs.google.com/spreadsheets/d/1Ffdzco_Bl0Q-eNLxBIokN2-x7qRitPnnbr0wUaA3YXI/edit

Role: Staging database used to store and structure AI-generated multiple-choice questions (30 MCQs per lesson).

Target Sheet/Tab: Unit 1 AI_Questions

- **Workflow Integrations:** Appended by Append row in sheet in Workflow 2 (2-create quiz.json).
    - Read by Get All Questions From Sheet in Workflow 3 (3- create google form.json).

Master Central Administration Sheet

Spreadsheet URL:
https://docs.google.com/spreadsheets/d/1g_Qks_6XJgpXt1uJ_NuJgiWXK4Cu06V8Wks3tN28orA/edit?gid=1332134792#gid=1332134792

Role: Relational database powering student records, exam tracking, and grade storage.

Sub-Sheets

- **student_data:** Master student directory containing full names and official email addresses. Read by Get students sheet in Workflow 4 (4- send exam to student.json).
- **form ID:** Exam registry table tracking generated exam titles (examName), Google Form IDs (formId), and dispatch statuses (ok/error). Appended by Workflow 3, read/updated by Workflow 4, and read by Workflow 5.
- **Degree:** Gradebook tracking log storing submission timestamps, scores, itemized responses, deduplication keys, and notification dispatch statuses. Appended hourly by Workflow 5, read and updated by Workflow 6.

3. Complete AI Prompt Reference

Prompt 1: Lesson Extraction & Summarization Prompt

Target Node: Basic LLM Chain (Workflow 1: 1-create doc.json)

Model Configuration: models/gemini-3.5-flash-lite

```
الدراسية المناهج تحليل في ومتخصص تعليمي خبير أنت.
وشاملة بدقة يلي ما منه واستخرج التالي التعليمي النص بتحليل قم:

1. lesson_number: الدرس رقم (أو وجد، إن الدرس/الوحدة رقم اكتب أو "محدد غير").
2. lesson_name: الأساسي الدرس اسم أو عنوان.
3. summary: للمحتوى ومبسط شامل ملخص.
4. main_concepts: الواردة الأساسية والمصطلحات بالمفاهيم قائمة.
5. learning_objectives: الدرس هذا من المتوقعة التعليمية بالأهداف قائمة.

كلام أو مقدمات أي دون { بـ وتنتهي } بـ تبدأ فقط نقي JSON بصيغة حصراً المخرجات تكون أن يجب :جداً هام تنبيه
توضيحي.

التعليمي المحتوى
```

Prompt 2: 30-MCQ Assessment Generation Prompt

Target Node: Basic LLM Chain (Workflow 2: 2-create quiz.json)

Model Configuration: models/gemini-3.5-flash-lite

```
You are an expert educational assessment generator.
```

Analyze the provided educational material, identify all the lessons or sections contained within it, and generate multiple-choice questions for each detected lesson.

QUESTION DISTRIBUTION PER LESSON:
- For each lesson found in the material, generate exactly 30 questions.
- Cover different concepts and sections of each lesson thoroughly.

QUESTION REQUIREMENTS:
- Each question must be based ONLY on the provided educational material.
- Do not introduce information that is not supported by the source.
- Each question must have exactly four options: A, B, C, and D.
- Exactly one option must be correct.
- The incorrect options must be plausible but clearly incorrect according to the source.
- Avoid duplicate or substantially repeated questions.
- Preserve the exact title, naming, and terminology used in the source material for each lesson.

OUTPUT FORMAT:
Return ONLY valid JSON.
Do not add Markdown (no ```json or ```).
Do not add explanations before or after the JSON.

Use exactly this JSON structure:

```
{
  "lessons": [
    {
      "lesson": "Exact Lesson Name and Number from the Source",
      "questions": [
        {
          "question": "...",
          "optionA": "...",
          "optionB": "...",
          "optionC": "...",
          "optionD": "...",
          "correctAnswer": "A"
        }
      ]
    }
  ]
}
```

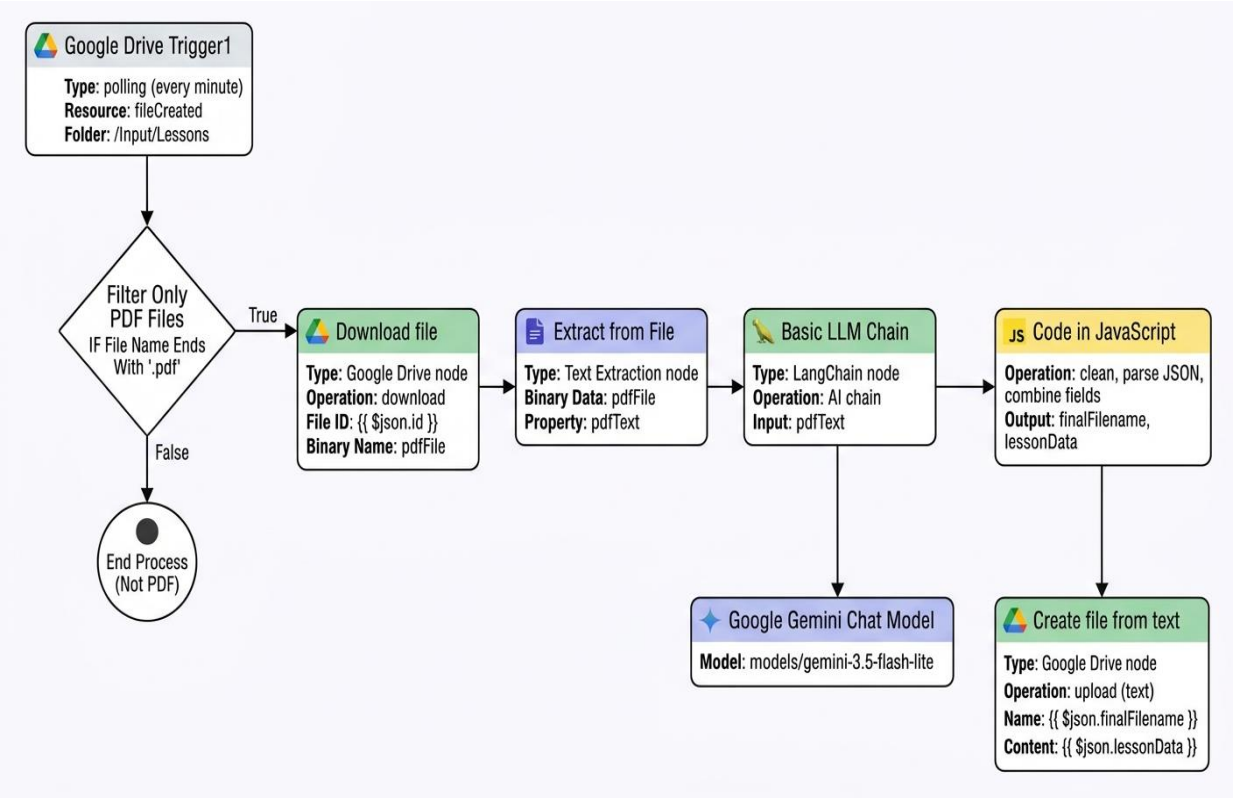
IMPORTANT VERIFICATION BEFORE OUTPUT:
- Each detected lesson must contain exactly 30 questions.
- Every question has optionA, optionB, optionC, optionD.
- Every correctAnswer is exactly one of A, B, C, or D.
- No question is missing any required field.

EDUCATIONAL MATERIAL

4. Detailed Workflow Specifications

Workflow 1: PDF Processing & Lesson Summary Extraction (1-create doc.json)

Monitors /PDF file in Google Drive. When a new PDF file is detected, it extracts binary text, utilizes Google Gemini AI (models/gemini-flash-lite) to extract structure, cleans and parses the JSON response, and writes a formatted summary file to /documentation.

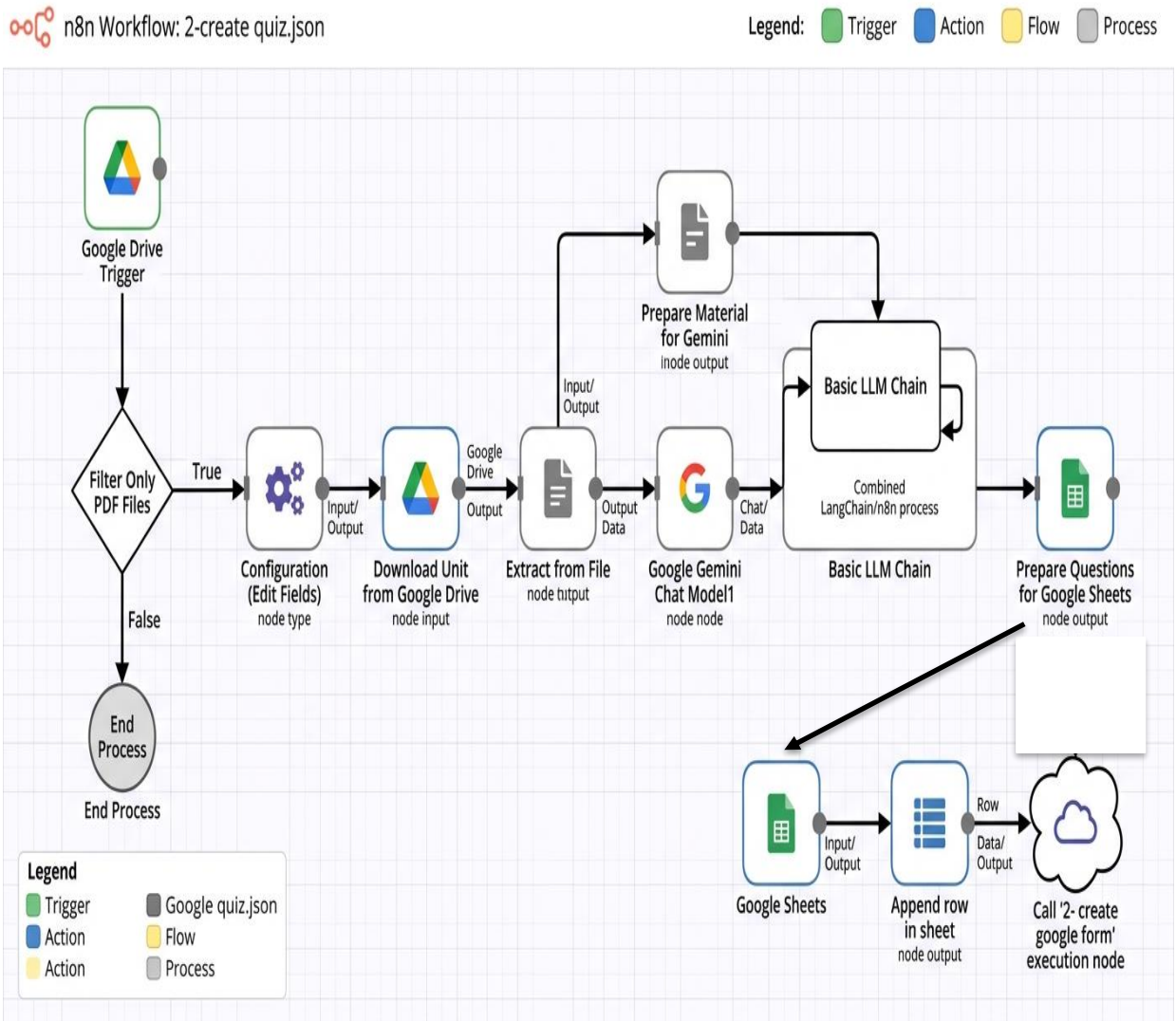


Step	Node Name	Connector / Type	Description & Functional Specification
1	Google Drive Trigger1	Google Drive Trigger	Periodically monitors /PDF file (every 1 min) for new uploads (file Created event).
2	Filter Only PDF Files	IF Conditional	Checks if incoming file name ends with ".pdf" to filter non-PDF uploads.
3	Download file	Google Drive Node	Fetches binary PDF content using the file ID (id) from the trigger payload.
4	Extract from File	Extract Text	Parses raw binary PDF into clean, machine-readable text string.
5	Basic LLM Chain	LangChain Chain	Combines extracted text with Prompt 1 to generate structured JSON output.

6	Google Gemini Chat Model	Gemini Model	Provides LLM backend using models/gemini-3-flash-lite.
7	Code in JavaScript	Code Node	Cleans raw AI response string, isolates JSON block, parses keys, and sets file naming metadata.
8	Create file from text	Google Drive Node	Writes structured text document containing title, summary, concepts, and objectives to /documentation.

Workflow 2: Quiz Generation (2-create quiz.json)

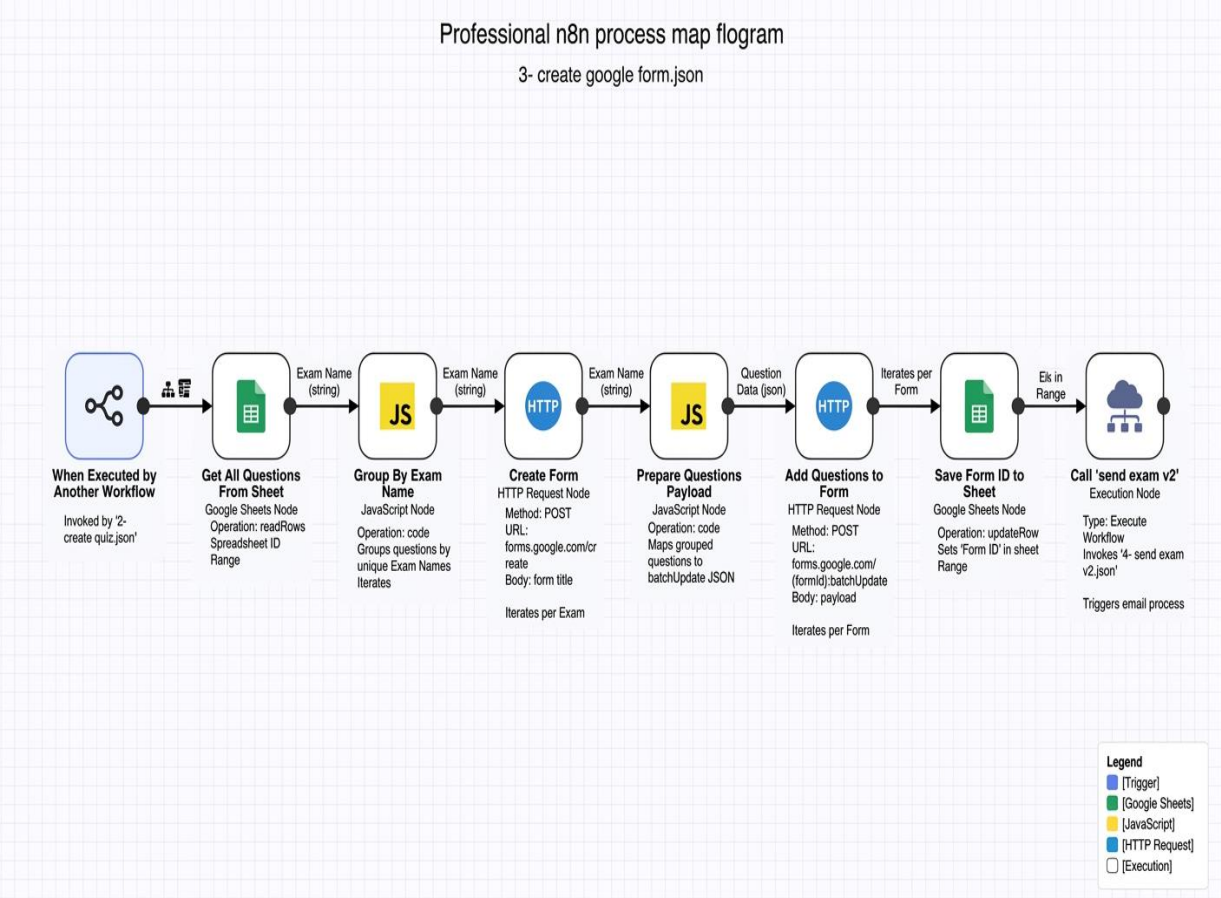
Monitors Google Drive for lesson documents, downloads and parses PDF content, uses Google Gemini AI to generate 30 MCQs per lesson in strict JSON format, formats questions into flat spreadsheet rows, appends them to Google Sheets, and triggers Workflow 3.



Step	Node Name	Connector / Type	Description & Functional Specification
1	Google Drive Trigger	Google Drive Trigger	Monitors folder for new file arrivals.
2	Filter Only PDF Files	IF Conditional	Validates that file extension matches .pdf.
3	Configuration	Set Node	Defines operational context: drive File ID, Spreadsheet ID, Sheet Name, Exam Name, Questions per Lesson, and Gemini Model parameters.
4	Download Unit from Google Drive	Google Drive Node	Downloads binary contents of target PDF using configured File ID.
5	Extract from File	Extract Text	Converts binary PDF into raw text string.
6	Prepare Material for Gemini	Code (JavaScript)	Formats raw text into custom context payload for the LLM chain.
7	Basic LLM Chain	LangChain Chain	Sends prompt and context to Gemini requesting 30 MCQs per lesson in strict JSON.
8	Google Gemini Chat Model1	Gemini Model	Configures execution model (models/gemini-flash-lite).
9	Prepare Questions for Google Sheets	Code (JavaScript)	Validates JSON structure, handles missing keys, and flattens array into row objects for spreadsheet Insertion (Question, Option A/B/C/D, Correct Answer, Exam Name).
10	Append row in sheet	Google Sheets Node	Appends structured question rows to "*Unit 1 AI_Questions*".
11	Call '2- create google form'	Execute Workflow	Triggers Workflow 3 (3- create google form.json).

Workflow 3: Google Form Creation (3- create google form.json)

Triggered by Workflow 2. Reads staged question rows from Google Sheets, groups items by *examName*, issues Google Forms API calls to construct a new form, provisions quiz parameters (*scoring, option keys, mandatory email capture*), logs the generated *formId* to the administration spreadsheet, and triggers Workflow 4.

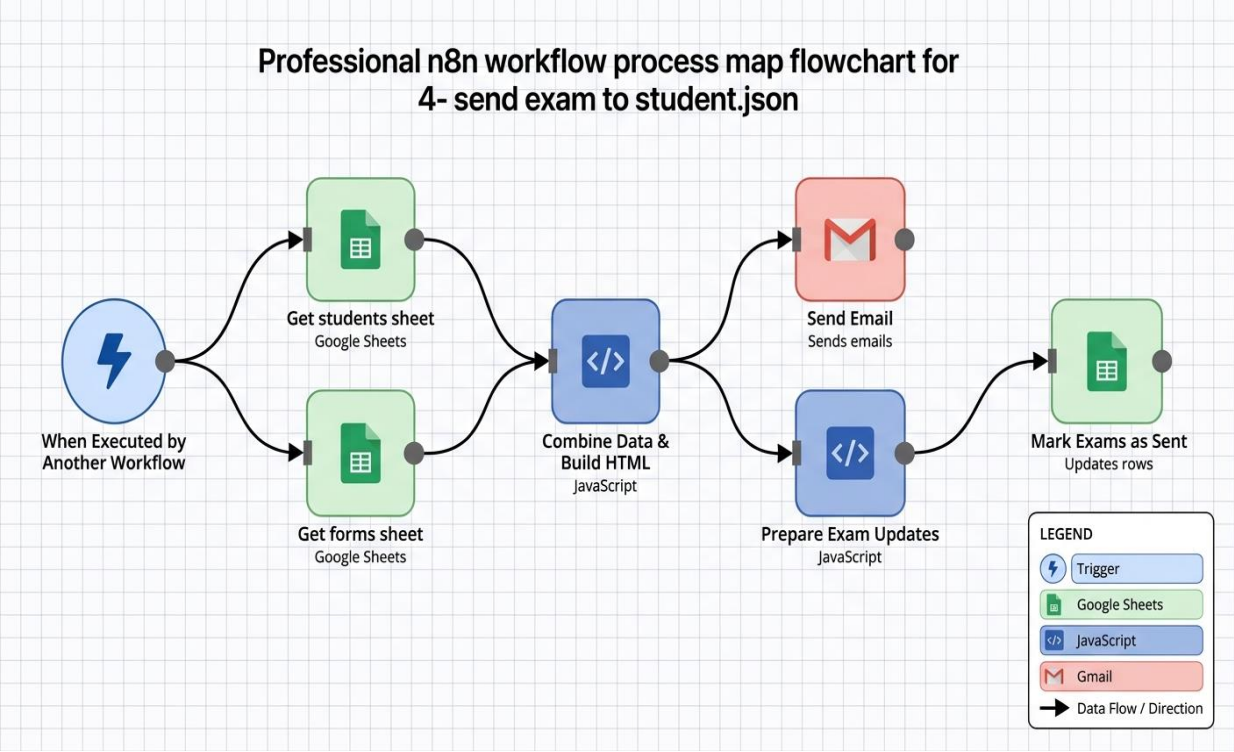


Step	Node Name	Connector / Type	Description & Functional Specification
1	When Executed by Another Workflow	Execute Workflow Trigger	Receives execution invocation and payload from Workflow 2.
2	Get All Questions From Sheet	Google Sheets Node	Fetches generated question rows from "*Unit 1 AI_Questions*".
3	Group By Exam Name	Code (JavaScript)	Sorts and aggregates questions into structured arrays keyed by *examName*.
4	Create Form	HTTP Request Node	Sends POST request to Google Forms API (*https://forms.googleapis.com/v1/forms*) to initialize form object with exam title.
5	Prepare Questions Payload	Code (JavaScript)	Constructs :batchUpdate JSON payload: enables quiz mode (isQuiz: true), adds compulsory email collection, formats options, maps correct answer keys, and assigns item point values.
6	Add Questions to Form	HTTP Request Node	Transmits batch update payload via POST to *https://forms.googleapis.com/v1/forms/{formId}*:batch Update.

7	Save Form ID to Sheet	Google Sheets Node	Logs "*examName*" and "*formId*" into the form ID registry sheet.
8	Call 'send exam v2'	Execute Workflow	Invokes Workflow 4 (4- send exam to student.json).

Workflow 4: Exam Email Dispatch (4- send exam to student.json)

Triggered by Workflow 3. Fetches student directory and active form registries, identifies pending exam dispatches, constructs personalized HTML email templates containing direct Google Form links, dispatches messages via Gmail API, and updates registry status flags to *ok*.

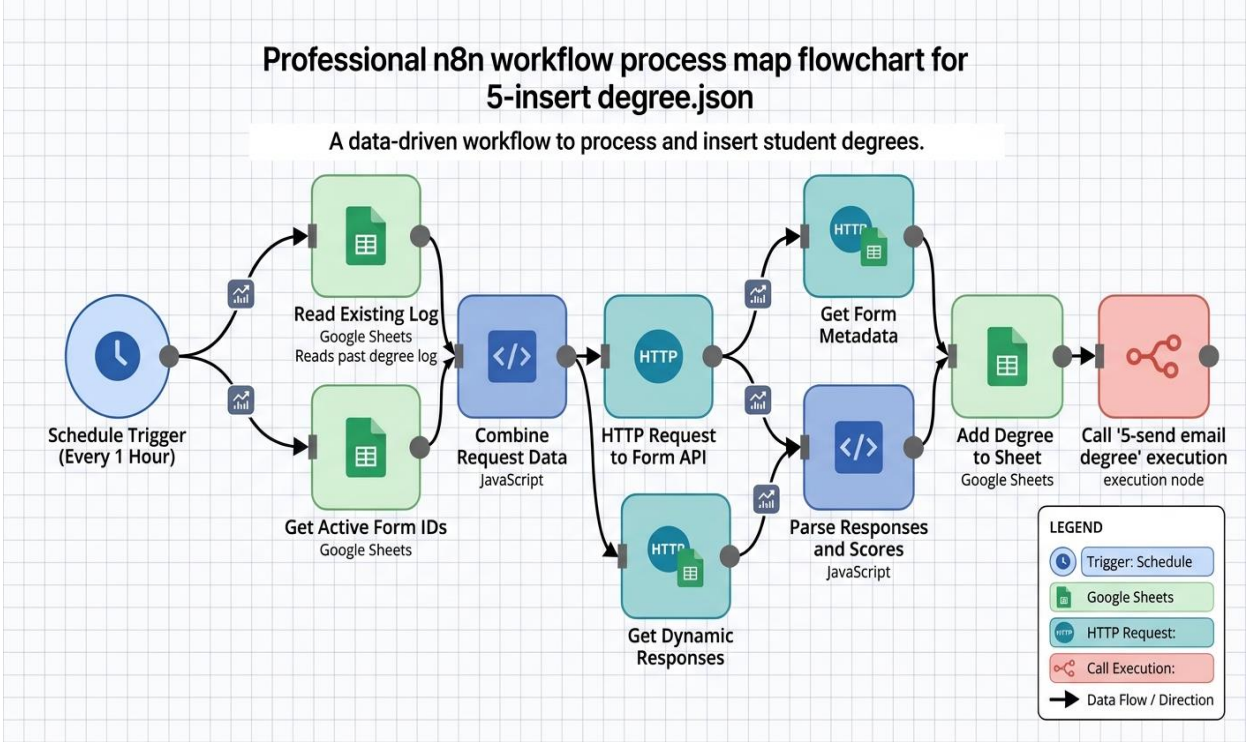


Step	Node Name	Connector / Type	Description & Functional Specification
1	When Executed by Another Workflow	Execute Workflow Trigger	Receives execution call from Workflow 3.
2	Get students sheet	Google Sheets Node	Fetches names and email addresses from "*student_data*" sheet.
3	Get forms sheet	Google Sheets Node	Fetches form registry records, IDs, and status flags from form ID.
4	Combine Data & Build HTML	Code (JavaScript)	Cross-references active students against unsent forms (status != ok), building customized HTML email payloads containing exam access URLs.

5	Send Email	Gmail Node	Dispatches HTML emails via Gmail API to target recipients.
6	Prepare Exam Updates	Code (JavaScript)	Formats array of successfully dispatched form IDs for status updating.
7	Mark Exams as Sent	Google Sheets Node	Updates form ID sheet, setting delivery status to ok for processed forms.

Workflow 5: Grade Processing & Insertion (5-insert degree.json)

Runs on a 1-hour schedule. Reads existing log entries to establish a deduplication registry, retrieves active form IDs, calls the Google Forms API to retrieve form schemas and submission records, parses and grades student answers, normalizes submission keys, appends new records to the Degree sheet, and triggers Workflow 6.

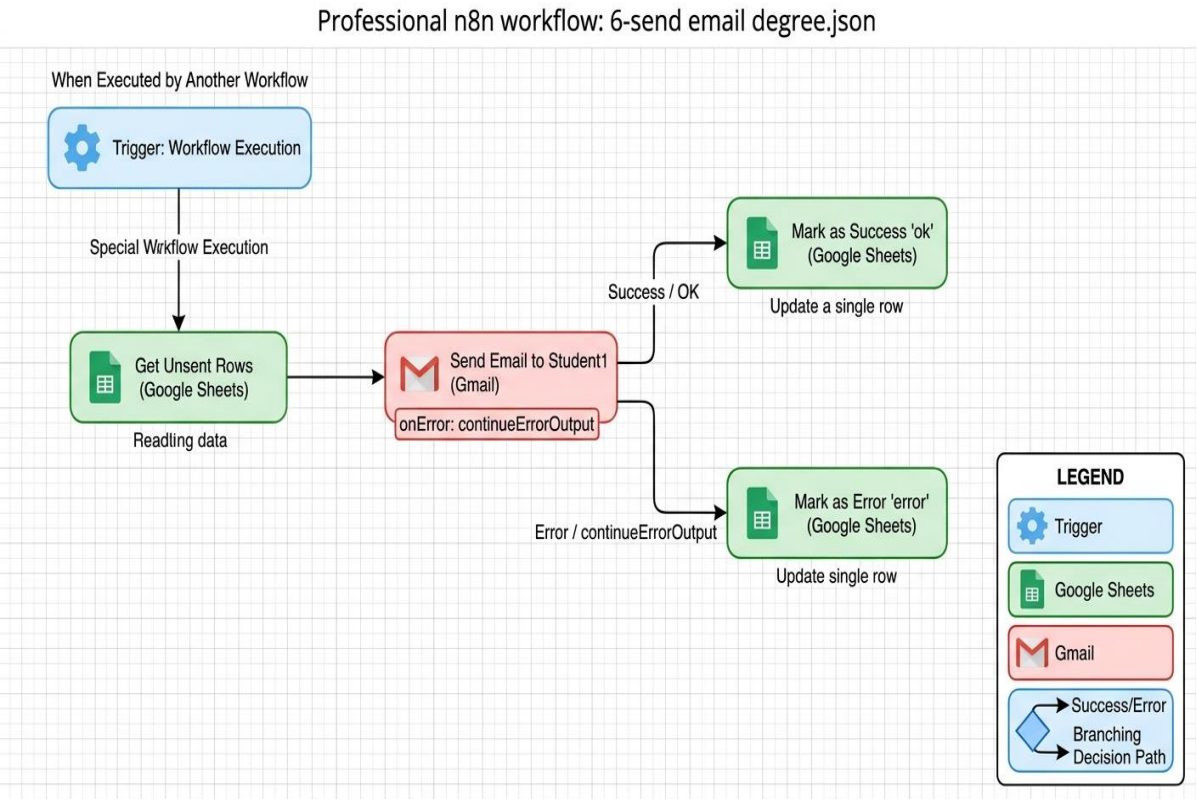


Step	Node Name	Connector / Type	Description & Functional Specification
1	Schedule Trigger (Every 1 Hour)	Schedule Trigger	Executes workflow automatically on a 60-minute recurring interval.
2	Read Existing Log1	Google Sheets Node	Reads historical records from Degree sheet to build deduplication set.
3	Get Active Form IDs1	Google Sheets Node	Retrieves active *formId* records from form ID registry.

4	Get Form Metadata1	HTTP Request Node	Queries "*https://forms.googleapis.com/v1/forms/{formId}*" to parse form item structure and locate email field IDs.
5	Get Dynamic Responses1	HTTP Request Node	Queries "*https://forms.googleapis.com/v1/forms/{formId}/responses*" to extract raw submission vectors and system grades.
6	Parse Responses & Scores1	Code (JavaScript)	Normalizes Arabic text, isolates submitter email, calculates total points, checks submission hash against existing logs to eliminate duplicates, and formats new grade rows.
7	add degree in sheet	Google Sheets Node	Appends new, unique student grade records into Degree sheet.
8	Call '5-send email degree'	Execute Workflow	Invokes Workflow 6 (6-send email degree.json).

Workflow 6: Grade Email Notification (6-send email degree.json)

Triggered by Workflow 5. Scans Degree sheet for pending grade notification records, dispatches score summary emails via Gmail, handles runtime errors with onError: continueErrorOutput, and updates row status flags to *ok* or *error* based on transmission results.



Step	Node Name	Connector / Type	Description & Functional Specification
1	When Executed by Another Workflow	Execute Workflow Trigger	Receives execution call from Workflow 5.
2	Get Unsent Rows	Google Sheets Node	Queries Degree sheet for rows where the sending status field is empty or pending.
3	Send Email to Student1	Gmail Node	Dispatches individual grade notifications. Configured with onError: continueErrorOutput to handle transmission failures gracefully without halting execution.
4	Mark as Success ('ok')1	Google Sheets Node	On successful transmission (Output 1), updates the corresponding row's status field to ok using row_number.
5	Mark as Error ('error')1	Google Sheets Node	On transmission failure (Output 2), updates the corresponding row's status field to error using row_number.

5. Technical Specifications: APIs, Authentication & Rate Limits

Workflow ID & Name	Node Name(s)	API Target Endpoint	Auth Method
WF 1: PDF Processing	Google Drive Trigger1	GET /drive/v3/files	OAuth2 (googleDriveOAuth2Api)
WF 1: PDF Processing	Download file	GET /drive/v3/files/{fileId}?alt=media	OAuth2 (googleDriveOAuth2Api)
WF 1: PDF Processing	Basic LLM Chain + Gemini Model	POST /v1beta/models/gemini-3.5-flash-lite:generateContent	API Key / Managed AI Gateway
WF 1: PDF Processing	Create file from text	POST /upload/drive/v3/files?uploadType=multipart	OAuth2 (googleDriveOAuth2Api)
WF 2: Quiz Generation	Google Drive Trigger	GET /drive/v3/files	OAuth2 (googleDriveOAuth2Api)
WF 2: Quiz Generation	Download Unit...	GET /drive/v3/files/{fileId}?alt=media	OAuth2 (googleDriveOAuth2Api)
WF 2: Quiz Generation	Basic LLM Chain + Gemini Model	POST /v1beta/models/gemini-3.5-flash-lite:generateContent	API Key / Managed AI Gateway
WF 2: Quiz Generation	Append row in sheet	POST /v4/spreadsheets/{spreadsheetId}/values/{range}:append	OAuth2 (googleSheetsOAuth2Api)

WF 3: Form Creation	Get All Questions...	GET /v4/spreadsheets/{spreadsheetId}/values/{range}	OAuth2 (googleSheetsOAuth2Api)
WF 3: Form Creation	Create Form	POST /v1/forms	Google OAuth2
WF 3: Form Creation	Add Questions to Form	POST /v1/forms/{formId}:batchUpdate	Google OAuth2
WF 3: Form Creation	Save Form ID to Sheet	POST /v4/spreadsheets/{spreadsheetId}/values/{range}:append	OAuth2 (googleSheetsOAuth2Api)
WF 4: Exam Dispatch	Get students sheet & Get forms sheet	GET /v4/spreadsheets/{spreadsheetId}/values/{range}	OAuth2 (googleSheetsOAuth2Api)
WF 4: Exam Dispatch	Send Email	POST /gmail/v1/users/me/messages/send	OAuth2 (gmailOAuth2)
WF 4: Exam Dispatch	Mark Exams as Sent	PUT /v4/spreadsheets/{spreadsheetId}/values/{range}	OAuth2 (googleSheetsOAuth2Api)
WF 5: Grade Processing	Read Existing Log1 & Get Active Form IDs1	GET /v4/spreadsheets/{spreadsheetId}/values/{range}	OAuth2 (googleSheetsOAuth2Api)
WF 5: Grade Processing	Get Form Metadata1	GET /v1/forms/{formId}	Google OAuth2
WF 5: Grade Processing	Get Dynamic Responses1	GET /v1/forms/{formId}/responses	Google OAuth2
WF 5: Grade Processing	add degree in sheet	POST /v4/spreadsheets/{spreadsheetId}/values/{range}:append	OAuth2 (googleSheetsOAuth2Api)
WF 6: Grade Notification	Get Unsent Rows	GET /v4/spreadsheets/{spreadsheetId}/values/{range}	OAuth2 (googleSheetsOAuth2Api)
WF 6: Grade Notification	Send Email to Student1	POST /gmail/v1/users/me/messages/send	OAuth2 (gmailOAuth2)
WF 6: Grade Notification	Mark as Success / Mark as Error	PUT /v4/spreadsheets/{spreadsheetId}/values/{range}	OAuth2 (googleSheetsOAuth2Api)

6. Maintenance, Operations & Troubleshooting Guide

Operational Execution Order & Invocation Topology

To ensure operational consistency across the automation suit, maintainers must understand the dual execution models used:

```
[ Event / Trigger Invocation ]
  ├── Workflow 1: Standalone (Google Drive Trigger1)
  ├── Workflow 2: Standalone (Google Drive Trigger)
  └── Workflow 5: Standalone (Schedule Trigger Every 1 Hour)

[ Chained Sequential Execution ]
  Workflow 2 (Call '2- create google form')
      └──► Workflow 3 (Call 'send exam v2')
              └──► Workflow 4 (Terminal)

  Workflow 5 (Call '5-send email degree')
      └──► Workflow 6 (Terminal)
```

- **Workflow 1:** Operates independently, monitoring Google Drive /PDF file to output text summaries into /documentation.
- **Workflow 2 -> Workflow 3 -> Workflow 4 Chain:** Triggered by new PDF uploads. Workflow 2 parses questions, stores them in Google Sheets, and explicitly calls Workflow 3. Workflow 3 builds the Google Form, updates the registry, and calls Workflow 4, which emails students.
- **Workflow 5 -> Workflow 6 Chain:** Runs on an automated schedule every 60 minutes. Workflow 5 processes student quiz submissions, appends scores to the Degree sheet, and triggers Workflow 6, which sends individual score notifications.

Failure Diagnostics & Recovery Procedures

- **Issue: Workflow 1 or 2 fails to trigger upon uploading a file:** Check if the uploaded file is in the root directory rather than /PDF file. Verify that Filter Only PDF Files is not evaluating to false due to file extension casing (.PDF vs .pdf). Confirm googleDriveOAuth2Api credentials.
- **Issue: Gemini AI generation failure or malformed JSON error:** Review output in Basic LLM Chain. If Gemini encounters rate limits (15 RPM), it may return an error HTTP status code or truncate output. Verify JavaScript cleanup node isolates JSON. Upgrade Gemini API tier if needed.
- **Issue: Workflow 3 fails during Add Questions to Form:** Examine the payload generated by Prepare Questions Payload. Ensure option arrays do not contain empty/null strings. Verify Google OAuth2 credentials have write permissions to the Google Forms scope.
- **Issue: Grade emails failing to send in Workflow 6:** Send Email to Student1 utilizes onError: continueErrorOutput. Check if execution routed to Mark as Error ('error')1. Verify Gmail API daily send limits have not been exceeded.